

Hulle_Madeline_Individual_Assignment_05

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Set Up

Packages

```
# general use and cleaning

library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```

# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

#survey data (reading different sheets in as seperate data frames)

aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information

sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quaity

water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  # values in df that are actually indicating missing values,
  # making it seem as a chr when it should be num
  # different ways missing values show up in data set
  na = c("over", "", "n/a", "N/A", "173+")
)

```

Class Code

2. Cleaning

sites object

```

# creating new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sites
  filter(sector == "NCOS" &
    sampling_frequency == "Quarterly")

```

clean taxon_list

```
# creating new object from taxon_list
taxon_list_clean <- taxon_list |>
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

clean water_quality

```
# creatign new object from water_quality
water_quality_clean <- water_quality |>
  # cleaning column name
  clean_names() |>
  # a filtering join, only keep the rows in wq clean in which the
  # site matches up with the site in ncos clean
  semi_join(ncos_sites, by = "site") |>
  # creating new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # grouping by date/site
  group_by(date_site) |>
  # calculating the medians
  summarize(
    med_pH = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungrouping df
  ungroup() |>
  # filtering out outliers
  filter(date_site != "2022-08-12_NVBR")
```

clean aq_ins

```
# create a new object
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join only include quarterly ncos surveys
  semi_join(ncos_sites, by = "site") |>
```

```

# renaming date_on_vial column to date
rename(date = date_on_vial) |>
# selecting columns of interest
# keeping information about each survey and the 9 most abundant taxa
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filtered to only include filteres beaker samples
# (better sampling technique)
filter(sample_type %in% c("FB 250", "FB250")) |>
# converting to long format, making df longer
pivot_longer(
  # name the columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for old column names
  # column names are all in new column called taxon
  names_to = "taxon",
  # new column for the values in each of the old columns
  # counts now under a column called abundance
  values_to = "abundance"
) |>
# group by date site and taxon
group_by(date, site, taxon) |>
# calculated average count per liter (total abundance/7.7)
summarise(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungrouping data frame
ungroup() |>
# new column to join data frames, didnt remove them
unite("date_site", date, site, remove = FALSE) |>
# joined with water quality clean
left_join(water_quality_clean, by = "date_site") |>
# joined with taxonomic information
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# fill in full site names
mutate(site_full = case_when(

```

```

site == "NVBR" ~ "North of Venoco Bridge",
site == "NEC" ~ "East Channel",
site == "NMC" ~ "Main Channel",
site == "NPB" ~ "South of Phelps Bridge",
site == "NPB1" ~ "North of Phelps Bridge",
site == "NPB2" ~ "Phelps Road",
site == "NWP" ~ "West Pond",
site == "NDC" ~ "Devereux Creek"
)) |>
# set factor levels for full site name by median salinity
mutate(site_full = fct_reorder(site_full,
                               med_sal,
                               .fun = "median",
                               .na_rm = TRUE),
       # reversing the order
       site_full = fct_rev(site_full))

```

Visualizing

Salinity

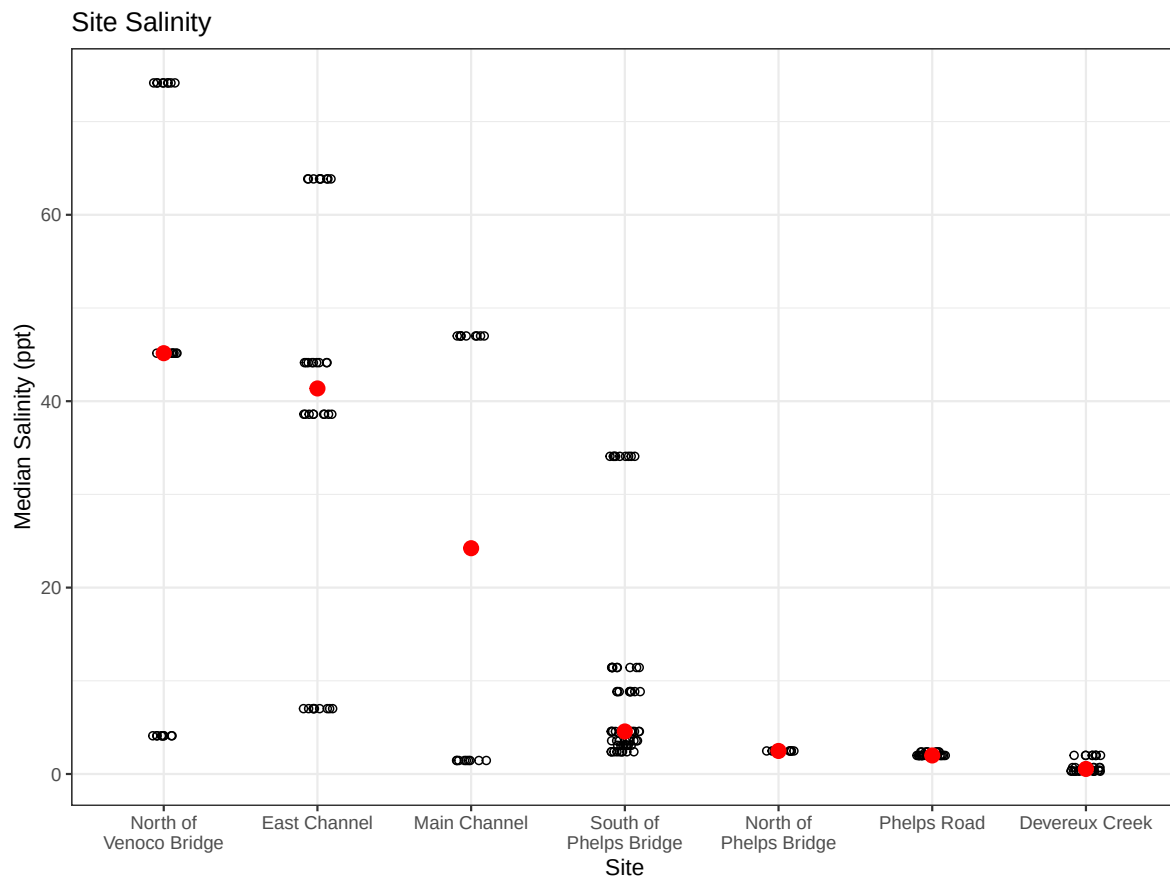
```

# creating new object to store plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal)) +
# points representing median salinity at each survey
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# points representing mean salinity
stat_summary(geom = "point",
            fun = median,
            color = "red",
            size = 3) +
# adding line breaks to x axis labels (wrapping)
scale_x_discrete(labels = label_wrap(width = 15)) +
# relabeling axes and adding a title
labs(x = "Site",
     y = "Median Salinity (ppt)",
     title = "Site Salinity") +

```

```
# adding themeing
theme_bw()

# visualizing the object
salinity_plot
```



Copepod Abundance

Cleaning

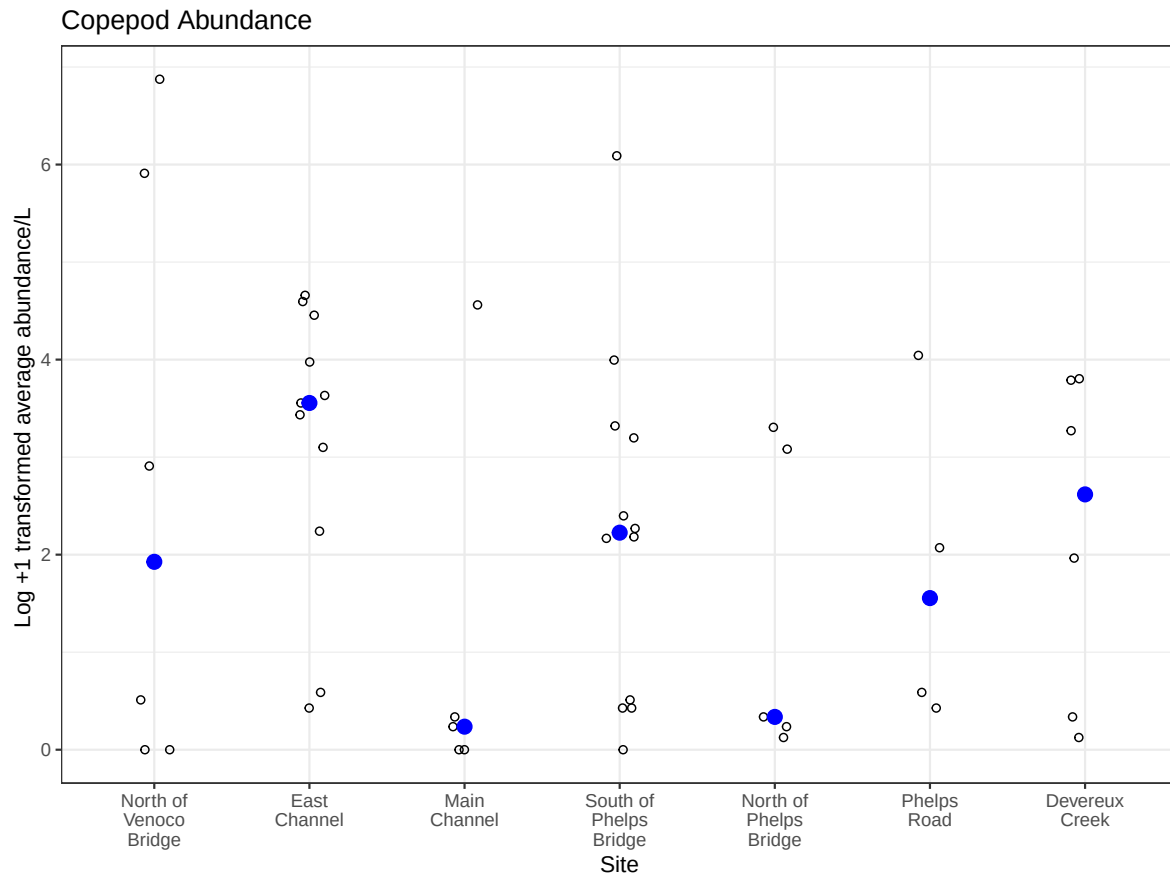
```
copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
  # adding 1 bc some observations are 0
  # helps overcome issues with large differences in order of magnitude
  mutate(log_ave_lit = log(ave_lit + 1))
```

Copepod Viz

```
# creating object to store plot
copepod_plot <- ggplot(data = copepods,
                      mapping = aes(x = site_full,
                                    # using log transformed count per liter
                                    y = log_ave_lit)) +

# jitter to represent each sample
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# points to represent median copepod abundance
stat_summary(geom = "point",
            fun = median,
            color = "blue",
            size = 3) +
# wrappign x-axis labels
scale_x_discrete(labels = label_wrap(width = 10)) +
# relabeling axes
labs(x = "Site",
     y = "Log +1 transformed average abundance/L ",
     title = "Copepod Abundance") +
# using different theme
theme_bw()

# displaying the object
copepod_plot
```



Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: `site_full`
- y-axis: `log_ave_lit` (updated w/ `log_ave_lit` after viz)
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary(geom = "point", fun = median)`

b. Wrangle your data

```
# creating a new object `ostracods` from `aq_ins_clean`
ostracods <- aq_ins_clean |>
  # filtering to only include ostracods
  filter(taxon == "ostracod") |>
  # log transforming `ave_lit` to overcome large difference in
  # order of magnitude between observations
  mutate(log_ave_lit = log(ave_lit + 1))

# taking a sample of 10 observations
slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
  date_site      date            site taxon ave_lit med_ph med_sal med_do
  <chr>          <dtm>          <chr> <chr>   <dbl>   <dbl>   <dbl>   <dbl>
1 2023-09-15_NMC 2023-09-15 00:00:00 NMC  ostr~    0.4    NA     NA     14.2
2 2022-08-10_NPB 2022-08-10 00:00:00 NPB  ostr~    2.67   8.28   8.84    9.81
3 2024-01-28_NPB 2024-01-28 00:00:00 NPB  ostr~    3.33   7.43   3.08    6.6
4 2023-05-05_NPB 2023-05-05 00:00:00 NPB  ostr~   24.7   NA     NA     12.1
5 2023-05-04_NEC 2023-05-04 00:00:00 NEC  ostr~    0.667  8.05   NA     8.55
6 2022-01-24_NEC 2022-01-24 00:00:00 NEC  ostr~    0.267 NA     NA     NA
7 2021-08-25_NPB 2021-08-25 00:00:00 NPB  ostr~  581.    7.52   4.57    4.5
8 2022-05-11_NDC 2022-05-11 00:00:00 NDC  ostr~    0     9.11   0.398   9.4
9 2022-05-13_NVBR 2022-05-13 00:00:00 NVBR ostr~    0     9.72  74.2   11.4
10 2023-05-04_NMC 2023-05-04 00:00:00 NMC  ostr~    1.87   8.32   NA     10.3
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code Your Figure

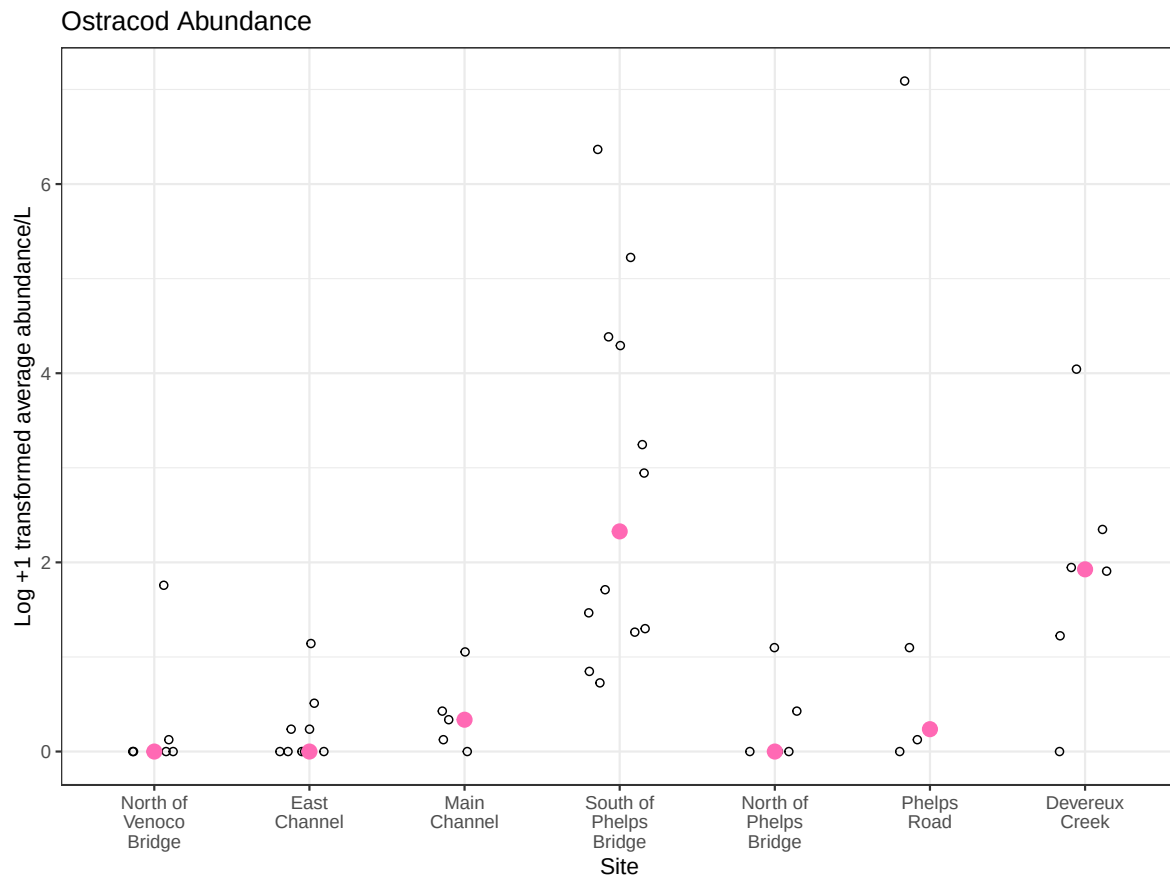
```
ostracod_plot <- ggplot(data = ostracods,
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
  geom_jitter(height = 0,
              width = 0.2,
              shape = 21) +
```

```

stat_summary(geom = "point",
             fun = median,
             color = "hotpink",
             size = 3) +
labs(x = "Site",
     y = "Log +1 transformed average abundance/L",
     title = "Ostracod Abundance") +
theme_bw() +
scale_x_discrete(labels = label_wrap(width = 10))

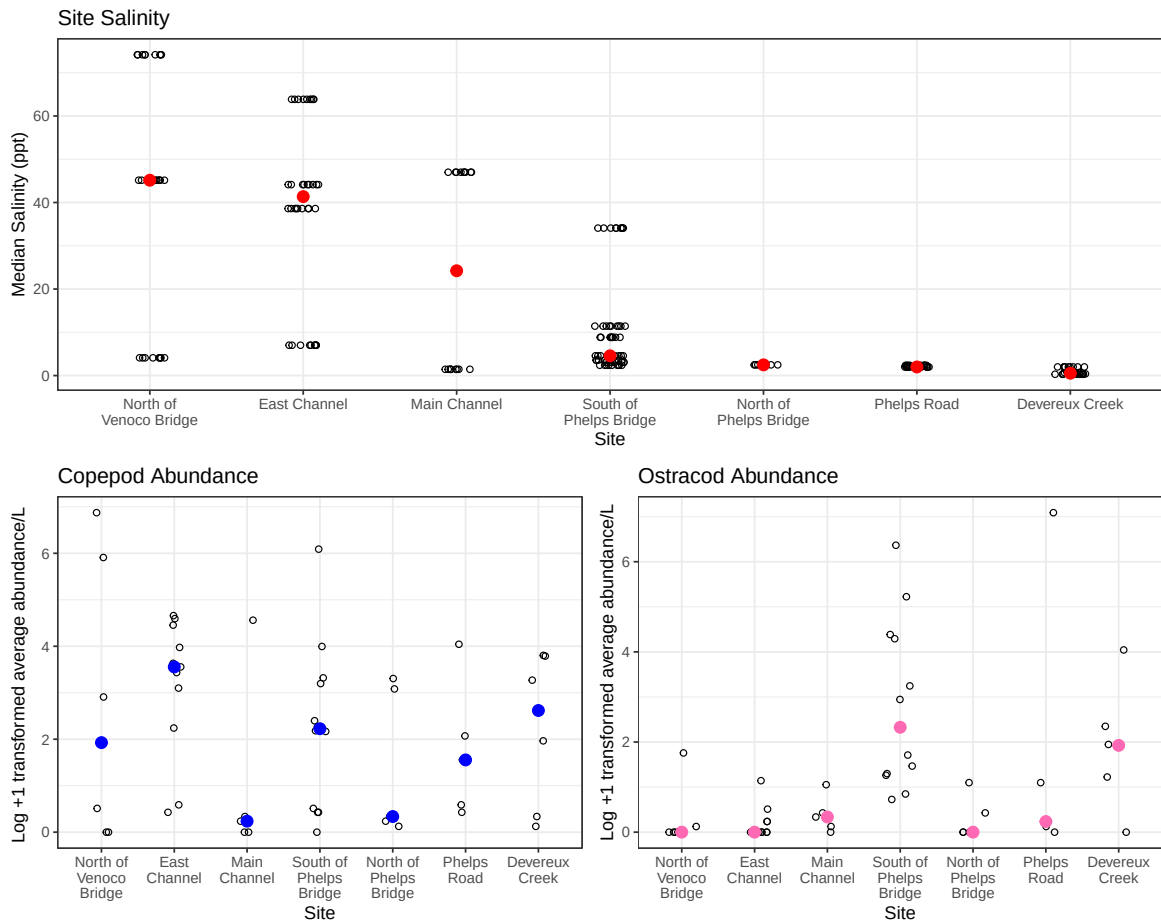
ostracod_plot

```



d. Create a Multi Panel Figure

```
salinity_plot / (copepod_plot | ostracod_plot)
```



e. Interpret Your Figure

- Between sites there does seem to be a difference between the median log + 1 average abundance/L of Copepods (blue dots) and that of Ostracods (pink dots). For sites North of Venoco Bridge and East Bridge, the median log + 1 average abundance/L of Ostracods is zero while for Copepods, this value is greater than zero, with East Channel being the site with the greatest median log + 1 average abundance/L of all of the sites. For the Main Channel site, Ostracod and Copepod median log + 1 average abundance/L are quite similar, both slightly above 0. The site South of Phelps Bridge also sees similar levels of

median log + 1 average abundance/L across both groups, with this being the highest median log + 1 average abundance/L seen for the Ostracods. At North of Phelps Bridge we see low median log + 1 average abundance/L for both groups, with Ostracod having a slightly lower median log + 1 average abundance/L than Copepods. Finally, Phelps Road and Devereux Creek see similar relative median log + 1 average abundance/L values across the two groups, but it is higher for Copepods at both sites. Overall, there seems to be, on average, higher Copepod abundance across these 7 sites than Ostracod abundance as more of the blue (median log + 1 average abundance/L for Copepods) points are higher up on the y-axis than the pink ones (median log + 1 average abundance/L for Ostracods).

- Comparing the Site Salinity plot to the Ostracod Abundance plot, there seems to be a possible relationship between the two. For site salinity, the mean salinity level is highest for North of Venoco Bridge and steadily decreases through the intermediate sites until the lowest median salinity is recorded at Devereux Creek. In the Ostracod Abundance plot, we see an almost inverse pattern where North of Venoco Bridge and East Channel see the lower median log + 1 average abundance/L levels and this steadily increases (with the exception of the South of Phelps Bridge observation) along the intermediate sites and reaching a peak at Devereux creek. More data would need to be collected and more analysis be done to determine if this is a true relationship and if the South of Phelps Bridge observation is a fluke) or if these inverse shapes are a coincidence in the data. From my visualization now however, it seems possible that higher salinity corresponds with a lower abundance of Ostracods and a lower salinity corresponds to a higher abundance of Ostracods. The optimal environmental conditions of Ostracods would also be helpful information to determine the validity of this relationship.